

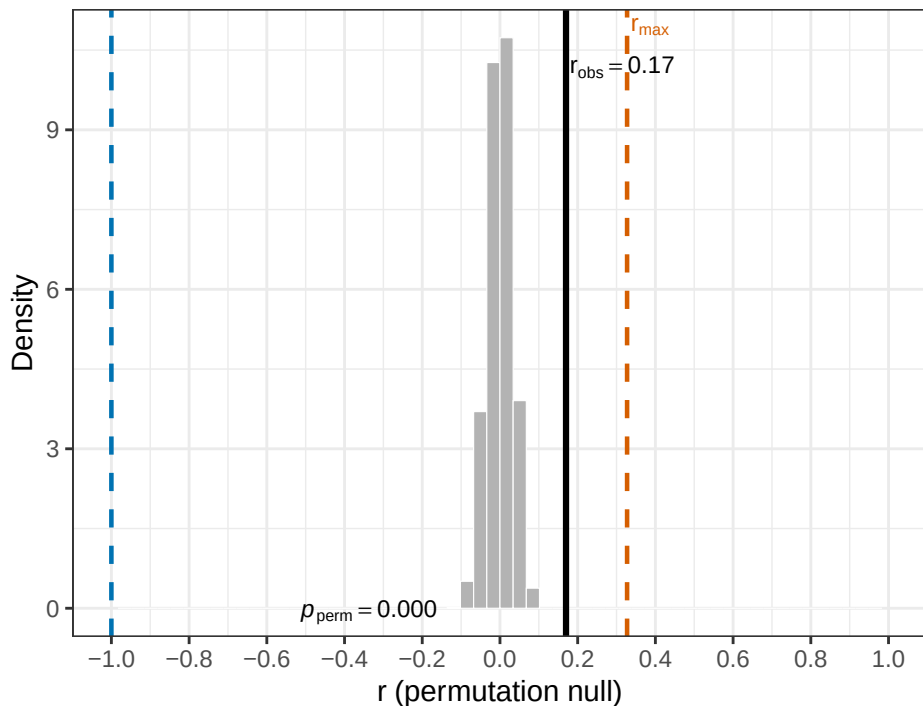
Permutation null distributions for two contrasting variable pairs

Vertical lines: $r_{\min}$ (blue dashed), $r_{\max}$ (orange dashed), r_{obs} (black solid).

Both pairs have similar r_{obs} and similar t-test p-values, yet r_{obs} occupies very different positions within the attainable range.

Pair A: Constrained marginals

$r_{\min} = -1$ $r_{\max} = 0.33$ $C_1 = 0.66$



Pair B: Unconstrained marginals

$r_{\min} = -0.99$ $r_{\max} = 1$ $C_1 = 0.99$

